

When Refining Math Instruction Data Hurts: A Controlled Study of Format Scaffolding and Verification Filtering

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Abstract

Rewriting and filtering instruction data before supervised fine-tuning (SFT) is now standard practice, and math is a favored target because answers are verifiable. We ask a simple question that the field rarely tests: *does aggressive math-data refinement actually generalize?* Using a strictly controlled protocol—identical base model (Qwen2.5-3B), identical LoRA configuration, identical evaluation, and *only the training data varies*—we decompose a representative refinement pipeline (**MISD**: structural typing $\rightarrow$ rule-based rewriting $\rightarrow$ symbolic verification) into its individual operators and measure each on an honest nine-benchmark suite of five in-distribution and four out-of-distribution (OOD) math benchmarks, including a Chinese benchmark (CMATH). We report three findings that contradict common assumptions. (1) The fully refined pipeline is the *worst* configuration on the honest suite (61.5 nine-benchmark average), below an unrefined Claude-rewrite baseline (63.7): in-distribution gains (+1.5) are erased by an OOD collapse driven by a cross-lingual format tax (CMATH -12.6). (2) The single largest harmful operator is the canonical “`\boxed{\}`” answer-format prompt, which helps in-distribution answer extraction (GSM8K +14) but destroys cross-lingual transfer (CMATH -30); the configuration that imposes *neither* the boxed prompt nor rewrite rules reaches 72.6 ± 0.4 (three seeds), +8.9 over the strongest baseline. (3) The symbolic-verification filter—a widely used math-data-cleaning step—contributes essentially nothing: dropping the same number of *random* samples matches dropping the solver-flagged samples (72.1 vs. 72.6). Our results argue that for math SFT data, format normalization is double-edged and verification filtering is overrated; lighter-touch refinement generalizes substantially better.

1 Introduction

Instruction tuning data quality strongly determines downstream model behaviour, and a large literature now “refines” SFT data by selecting high-value examples (Li et al. 2023, 2024; Liu et al. 2023) or rewriting them into cleaner forms (Xu et al. 2023). Mathematics is an especially attractive domain for refinement because answers are *verifiable*: one can rewrite a problem, re-solve both versions, and keep only examples whose answers agree (Lu et al. 2024). This has produced a standard math-data recipe—*structure the problem, rewrite it into a canonical format with an explicit answer marker, and filter by symbolic answer equivalence*.

We take this recipe seriously and test whether it generalizes. The key methodological move is a *strictly controlled* comparison: we fix the base model, the LoRA hyperparameters, the optimizer, and the evaluation protocol, and vary *only the training data*. This lets us attribute any accuracy change causally to a single data operator. We further insist on an *honest* evaluation: in addition to five in-distribution benchmarks whose format matches the (English, boxed) training distribution, we evaluate on four out-of-distribution benchmarks—a table-reasoning set, an adversarial-number set, and crucially a *Chinese* elementary-math set (CMATH)—that the refinement recipe never sees.

Under this lens the standard recipe looks much worse than its in-distribution numbers suggest. The fully refined pipeline wins in-distribution but is the *worst* configuration once OOD generalization is counted, because the very interventions that raise in-distribution accuracy (an English boxed-answer prompt and English rewrite templates) teach the model a rigid output format that fails to transfer. Removing those interventions recovers a large, statistically robust gain. And the symbolic-verification filter, often treated as the principled core of math-data cleaning, turns out to be statistically indistinguishable from dropping random examples.

Contributions.

- A confounder-controlled decomposition of a representative math-data refinement pipeline (structure / rewrite / verify) into its individual operators, each measured on an honest in-distribution + OOD suite (Section 4).
- The finding of a *format-scaffolding generalization tax*: the boxed-answer prompt and English rewrite rules trade in-distribution extraction for a large cross-lingual/OOD loss, making the full pipeline net-negative on the honest suite (Tables 2, 3).
- A difficulty-matched control showing that the symbolic-verification filter contributes ≈ 0 : random-sample dropping matches solver-flagged dropping (Section 4.4).
- Actionable guidance for practitioners: prefer light-touch refinement; do not impose a fixed answer-format scaffold if cross-lingual or OOD robustness matters; do not assume verification filtering adds value without a random-drop control.

2 Related Work

Instruction-data selection. LIMA (Zhou et al. 2023), IFD/Cherry (Li et al. 2023), SuperFiltering (Li et al. 2024), and DEITA (Liu et al. 2023) score and subselect general instruction data. These are domain-agnostic; we include them as baselines and show they transfer poorly to math. **Math-data synthesis and refinement.** MetaMath (Yu et al. 2023), WizardMath/Evol-Instruct (Xu et al. 2023), and MathGenie (Lu et al. 2024) rewrite or augment math problems, frequently using symbolic/solver checks to filter. Our work does not propose a new such method; it *stress-tests* the operators these methods share. **Less-is-more.** LIMO (Ye et al. 2025) and s1 (Muennighoff et al. 2025) argue small curated sets suffice for reasoning. Our minimal-refinement result is convergent evidence, but our mechanism is different: we attribute the effect to removing a format tax rather than to data scale. **Format and verification.** Boxed-answer extraction originates in MATH/Minerva-style evaluation (Hendrycks et al. 2021); SymPy answer-equivalence is a common cleaning primitive. We are not aware of prior work isolating their *OOD cost* under a controlled, single-variable design.

3 The Refinement Operators We Study

We instantiate the standard recipe as a three-stage pipeline and treat each stage as a toggleable operator.

Structure (typing). Each raw problem is typed along six fields {setup, conditions, question kind, format constraint, visual `asy`, MCQ} by Qwen2.5-7B-Instruct cross-checked with regex. Typing gates which rewrite rules may fire.

Rewrite (additive rules). Five *additive* rules (content may only be added, never deleted): R1 visual-label binding, R2 unit clarification, **R3 boxed-answer prompt** (append “*Give the final answer inside \boxed{}*”), R4 integer-type declaration, R5 long-LaTeX intermediate variable. R3 is by far the most frequently triggered ($\sim 82\%$).

Verify (two layers). Layer 1 is five pure-Python safety rules (S1–S5) that roll back any edit breaking LaTeX, variables, numbers, format constraints, or MCQ options. Layer 2 is a SymPy answer-equivalence filter: solve raw and rewritten versions with Qwen2.5-7B-Instruct, extract boxed answers, drop the example if they disagree. On our pool this drops 306 of 4,998 examples.

Each operator can be switched on or off independently, yielding the configurations compared below. We emphasize that the pipeline is the *instrument* of a controlled study, not the contribution.

4 Experiments

Setup. Base model Qwen2.5-3B (base). LoRA $r=16$, $\alpha=32$, $\text{lr } 1 \times 10^{-4}$, batch 4, grad-accum 8, 3 epochs, max length 2048—*identical for every row*. Inference via vLLM, temperature 0, boxed extraction with numeric fallback. Pool: 4,998 English math instructions.

Benchmarks. *In-distribution* (5): SVAMP, ASDiv, AddSub, MATH500, GSM8K. *Out-of-distribution* (4): MAWPS, TabMWP (table reasoning), GSM-Hard (adversarial numbers), CMATH (Chinese). The nine-benchmark average weights all benchmarks equally.

Significance. Multi-seed (3 training seeds) for the headline configurations; paired McNemar per benchmark.

4.1 Math needs domain-specific data, but selection is not enough

Table 1 confirms the usual in-distribution story: general-purpose selection methods (LIMA/IFD/SuperFiltering/DEITA/Cherry) collapse on math (50–55 avg), while LLM-rewrite refinement (Sanity, MISD) reaches ~ 75 . If we stopped here we would conclude refinement is a clear win—which is exactly the trap the next section springs.

Table 1: In-distribution accuracy (%) on five math benchmarks, Qwen2.5-3B. General selection baselines transfer poorly to math; LLM-rewrite refinement looks strongly positive *in-distribution*.

Method	Data	In-dist. Avg
<i>General-purpose selection (reproduced)</i>		
IFD	4.0k	50.22
SuperFiltering	4.0k	52.04
Random	4.0k	53.05
LIMA-style	4.0k	53.10
Cherry	4.0k	53.74
DEITA	4.0k	55.26
<i>LLM-rewrite refinement</i>		
Raw + boxed	5.0k	61.81
Sanity (Claude rewrite)	5.0k	73.79
Full MISD (V2)	4.7k	75.33

4.2 The honest suite: refinement is net-negative

Table 2 adds the four OOD benchmarks. The picture inverts: **Full MISD is the worst configuration** on the nine-benchmark average (61.5), *below* the unrefined Sanity baseline (63.7), because its +1.5 in-distribution edge is overwhelmed by a -6 OOD drop concentrated in CMATH (-12.6). The configuration that applies *neither* the boxed prompt nor the rewrite rules—just the clean rewrite—reaches **72.6**, +8.9 over the best baseline, with a tight three-seed spread.

4.3 The boxed prompt is the dominant harmful operator

Table 3 adds operators one at a time. The decomposition is monotone and striking: the SymPy filter is roughly neutral, **the boxed prompt costs -9.5** on the honest suite, and the remaining rules cost a further -1.2 . The boxed prompt—identified as “the only useful rule” by in-distribution ablation—is in fact the single most harmful intervention once OOD is counted: it lifts GSM8K extraction (+14) but collapses CMATH ($77 \rightarrow 47$) and MAWPS ($87 \rightarrow 50$).

4.4 The verification filter contributes nothing

Is the SymPy filter doing useful quality control, or merely deleting hard examples its 7B solver cannot handle? We run

Table 2: Honest evaluation: in-distribution (5) + OOD (4) math benchmarks, Qwen2.5-3B. “Box” = R3 boxed-answer prompt; “Rules” = R1–R5 rewrite; “Filt” = SymPy verification filter. Headline rows are mean $\pm$ std over 3 seeds. **Green**=best, **red**=worst on the nine-benchmark average.

Configuration	Box	Rules	Filt	In-dist. (5)	OOD (4)	9-bench Avg	CMATH
Sanity (Claude rewrite + boxed)	✓	✗	✗	73.79	51.07	63.69	47.74
Full MISD (V2)	✓	✓	✓	75.33	44.88	61.48	35.15
+ boxed only (R3, filtered)	✓	R3	✓	75.33	47.49	62.96	47.09
Minimal (no box, no rules)	✗	✗	✓	74.97	69.75	72.65	77.38
<i>random-drop control</i>	✗	✗	rand	74.76	68.73	72.08	75.38

Table 3: Operator ladder (9-bench avg). The much-touted boxed prompt is the largest *negative* contributor on the honest suite.

Configuration (cumulative)	9-bench	Δ
Sanity baseline (+ boxed)	63.69	—
Minimal: remove boxed & rules	72.65	+8.96
add back boxed (R3)	62.96	−9.69
add back all rewrite rules (= Full MISD)	61.48	−1.48

a *difficulty-matched control*: drop the *same number* (306) of *randomly chosen* examples instead of the solver-flagged ones, holding everything else fixed. The random-drop model reaches 72.08 vs. the SymPy-filtered 72.65 (Table 2)—statistically indistinguishable. The filter’s *choice* of which examples to drop adds no measurable value; the gain attributed to “verification” in in-distribution ablations is an artifact of (i) removing format scaffolding and (ii) difficulty-biased pruning, not of answer-equivalence checking.

4.5 Error analysis

The gap between configurations is concentrated in *answer extraction*, not reasoning: comparing correct/extraction-failure/computation-error rates, the boxed prompt mainly changes whether the model emits an extractable answer in the expected format, while computation-error rates are nearly constant across configurations. This is consistent with the format-tax interpretation: refinement teaches format compliance, which helps where the format matches and hurts where it does not.

5 Limitations

Single base scale. Headline results are on Qwen2.5-3B; we are extending to 7B/1.5B to test whether the format tax shrinks as base models already emit boxed answers by default (in progress). **Single source corpus.** All results use one 5,000-problem English pool; a second source corpus is needed to fully de-confound pipeline from corpus. **Solver-bound filter.** The verification filter uses a 7B solver, whose own accuracy bounds filter reliability; a stronger solver might recover value the random-drop control did not detect. We report these openly rather than over-claim.

6 Conclusion

A strictly controlled, honest evaluation overturns the in-distribution story of math instruction-data refinement. Rewriting problems into a canonical boxed English format raises in-distribution accuracy but imposes a large cross-lingual/OOD generalization tax, making the fully refined pipeline the worst configuration on an honest nine-benchmark suite; the symbolic-verification filter adds nothing beyond random pruning. The actionable message is to refine lightly: do not bake in a fixed answer-format scaffold when robustness matters, and validate any verification filter against a random-drop control.

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